

WHITEPAPER

AI-Assisted FinOps Report Response Portal

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Purpose

This document describes a proposed design for a controlled, AI-assisted FinOps report response workflow suitable for a highly locked-down enterprise AWS environment.

The proposed system allows teams to respond to financial report items through a secure, linked chat interface. The assistant helps users understand the report item, provide a justification, challenge the assignment, or request reassignment to a different technology catalogue owner. The system captures structured responses and publishes approved response data to S3 for dashboarding.

The design intentionally avoids broad AWS access, autonomous remediation, direct infrastructure changes, and direct technology catalogue mutation.

Executive Summary

The proposed solution is an **AI-assisted FinOps Report Response Portal**.

Each FinOps report item generates a secure deep link in an email notification. The user clicks the link, authenticates via enterprise SSO, and is taken to a case-specific chat interface. The AI assistant explains the report item using a narrow, pre-built case file, asks guided questions, and helps the user submit one of several controlled responses:

- Provide a justification.
- Challenge the financial allocation.
- Request reassignment to a different product, application, team, or technology catalogue entry.
- Mark the issue as needing further investigation.
- Attach supporting evidence.

The backend validates all actions, stores a structured response, writes an audit trail, and publishes dashboard-ready data to S3.

The model does **not** receive broad AWS service permissions. Bedrock is used only as a controlled reasoning and drafting layer. The application backend owns all reads, writes, validation, and authorisation.

Recommended Internal Name

Suggested name:

FinOps Response Portal

Alternative names:

- FinOps Justification Portal
- FinOps Assignment Validation Workflow
- AI-Assisted FinOps Response Workflow
- Cloud Cost Response Portal

Avoid names such as:

- Autonomous FinOps Agent
- AI Cost Remediation Agent
- Bedrock S3 Agent

Those names imply more autonomy and risk than the design requires.

Problem Statement

Enterprise FinOps reporting often identifies cost anomalies, allocation issues, unexplained variances, and ownership problems. However, the response process is commonly fragmented across email threads, spreadsheets, meetings, and manual dashboard updates.

This creates several problems:

Problem	Impact
Justifications are captured inconsistently	Dashboard commentary becomes stale or incomplete
Ownership challenges happen via email	No clear audit trail or structured reassignment workflow
Teams lack context when receiving reports	More time spent asking FinOps to explain the issue
Finance and engineering use different language	Poor translation between technical cause and financial impact
Responses are difficult to aggregate	Hard to show status, acceptance, disputes, and remediation progress
Manual chasing is required	FinOps team spends time coordinating instead of analysing

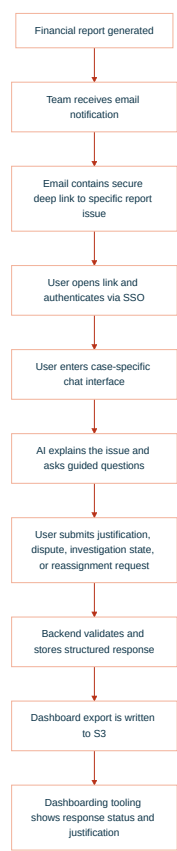
Proposed Solution

The solution is a controlled web-based workflow that combines:

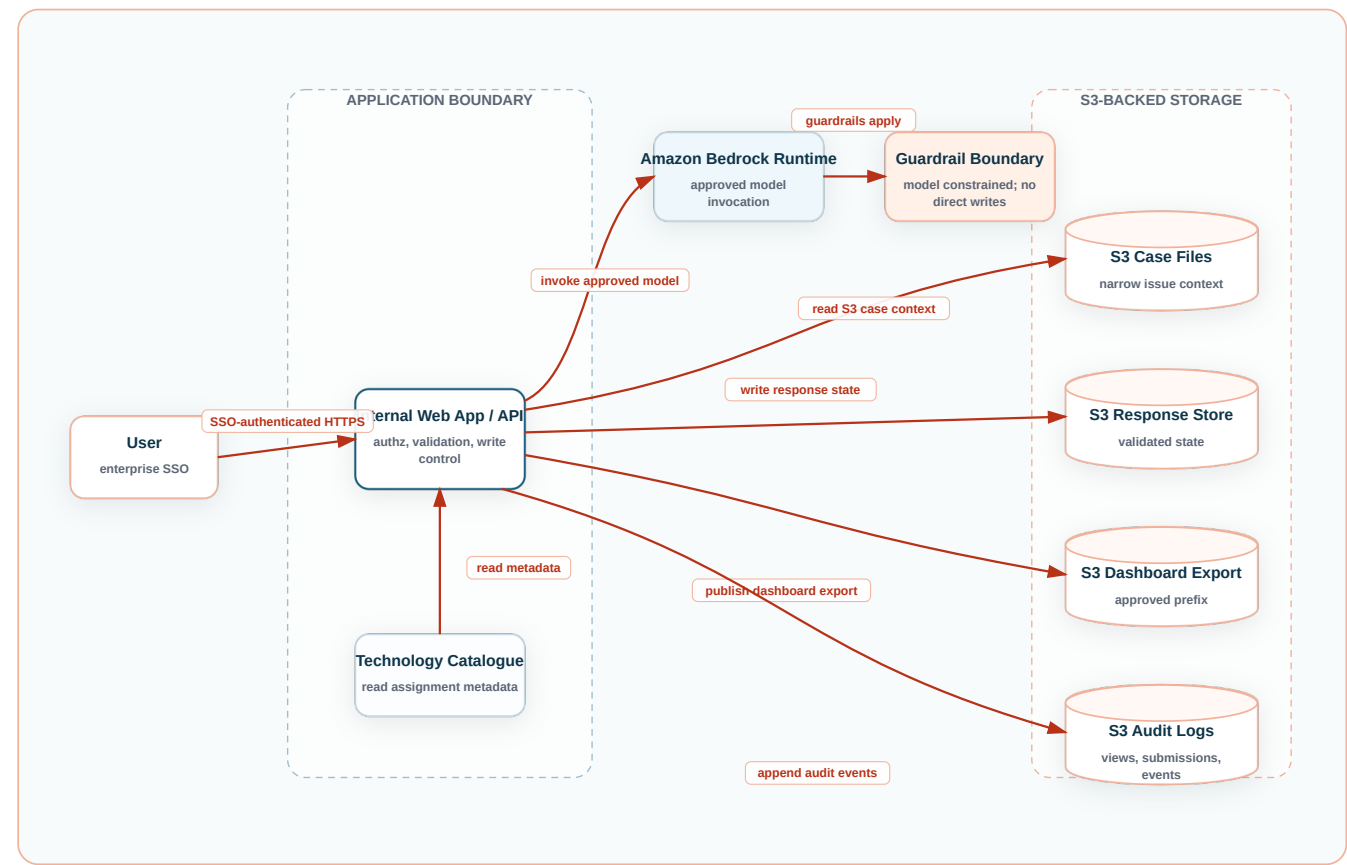
- A secure email deep link for each report item.
- Enterprise SSO authentication.
- Case-specific AI-assisted chat.
- Structured response capture.
- Optional reassignment request workflow.
- S3 dashboard export.
- Full audit logging.

The AI assistant is not a general-purpose chatbot. It is scoped to the specific report item and allowed actions for the authenticated user.

High-Level User Journey

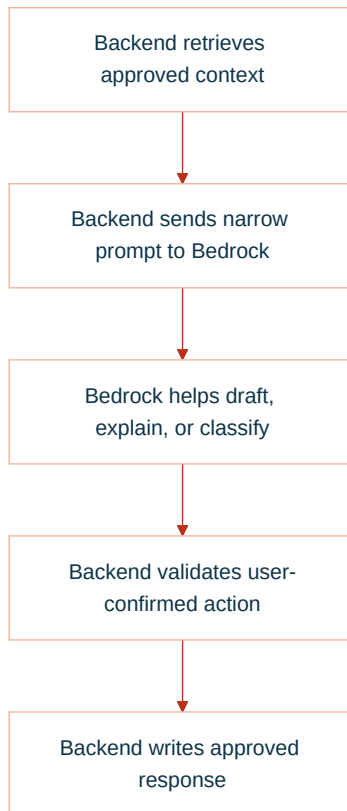


Security-Centred Architecture View

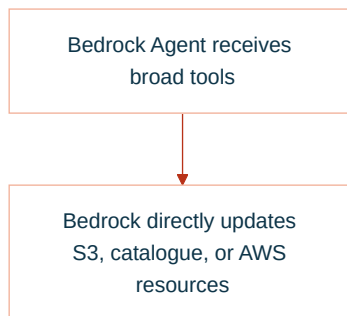


Key Design Principle

- The application backend performs all reads and writes.
- Bedrock receives only a narrow case file and returns assisted text, classifications, questions, or structured suggestions.
- The backend validates the user, the issue, the requested action, and the final response schema before writing anything.
- The design should be:



It should not be:



Scope of the MVP

In Scope

Capability	Description
Email deep link	Link users to a specific report issue
Enterprise SSO	Authenticate users before access
Case-specific chat	Chat only about the linked report item
Issue explanation	AI explains the cost movement and current assignment
Guided justification	AI helps users provide a structured response
Reassignment request	User can suggest a different product/application/team
Investigation state	User can mark the item as needing investigation
S3 dashboard export	Structured response written for dashboard consumption
Audit logging	Record who did what, when, and against which issue
Role-based access	Users see and update only authorised issues

Out of Scope for MVP

Not Required	Reason
Direct access to raw CUR	Upstream report process should create a case file
Access to EC2, RDS, EKS, or other live infra APIs	App is not inspecting or modifying infrastructure
IAM mutation	App does not manage permissions
Direct technology catalogue updates	Only reassignment requests are created
Autonomous remediation	Human-led response workflow only
Broad natural language access to all cost data	Case-specific interaction only
Public internet egress	Private endpoints and approved internal routes only
Static AWS access keys	SSO and IAM roles only
Broad Bedrock model access	One approved model for MVP
Cross-team visibility by default	Enforced RBAC/ABAC
Browser-based direct S3 writes	Backend-only writes

Core Response Types

Response Type	Description	Output
Justification	Team accepts the cost and explains why it occurred	Structured justification
Dispute	Team challenges the data, amount, or allocation	Dispute record
Reassignment request	Team believes another product/application/team owns the cost	Pending reassignment request
Needs investigation	Team cannot yet explain the item	Pending investigation status
Evidence attachment	Team adds supporting links or references	Evidence metadata

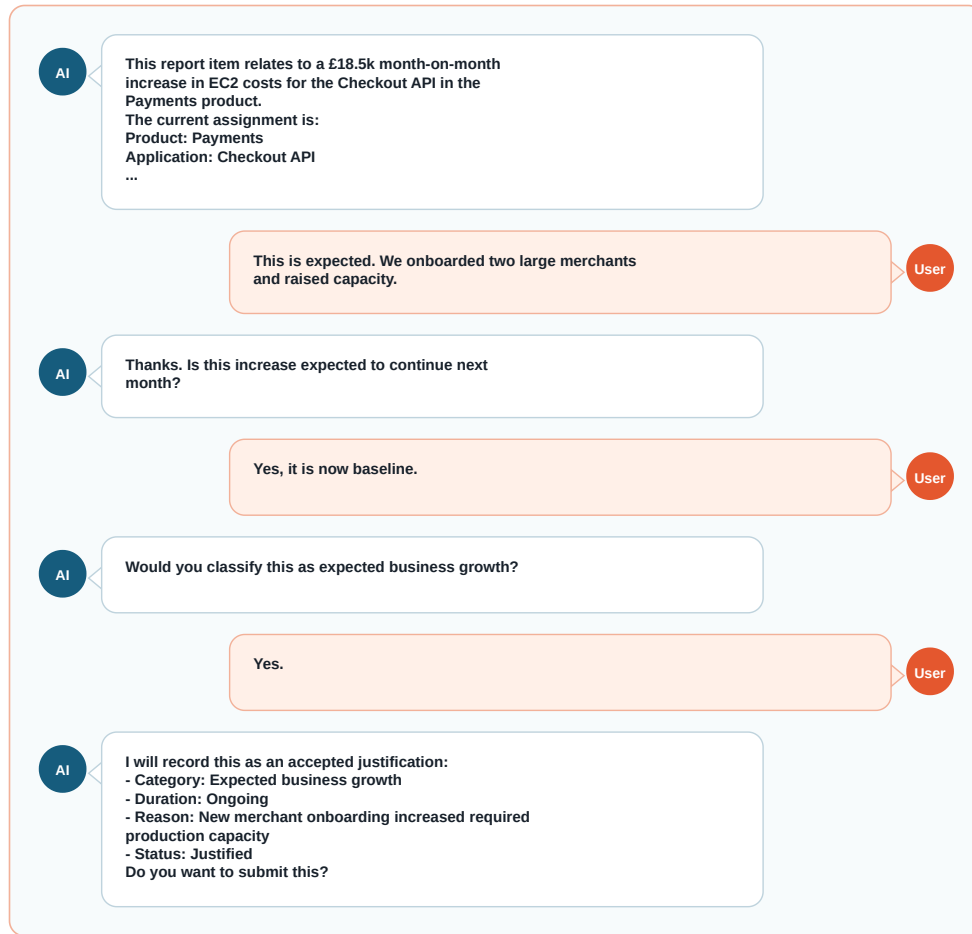
Suggested Status Model

Status	Meaning
awaiting_response	Team has not responded
in_discussion	Chat has started but no final response has been submitted
justified	Team accepted and justified the cost
disputed	Team challenges the data or allocation
reassignment_requested	Team proposed a different owner
pending_finops_review	Requires FinOps review
pending_catalog_update	Requires technology catalogue review/update
closed	No further action required
expired	No response received by due date

Suggested Justification Categories

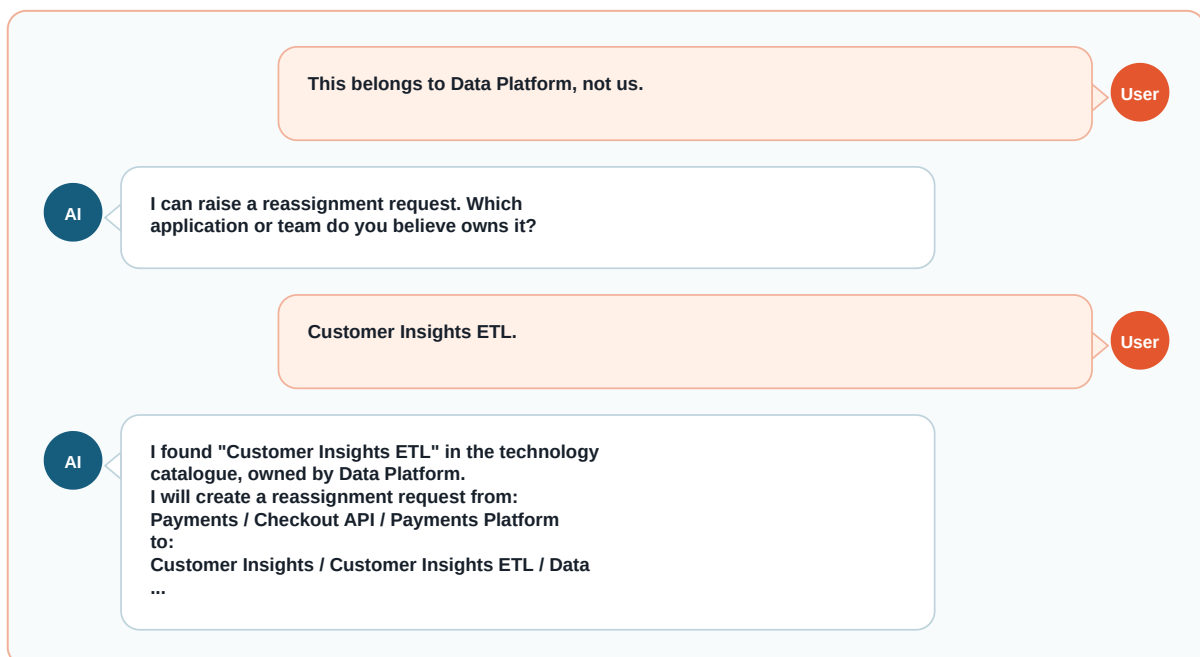
Category	Description
expected_business_growth	Spend increased due to genuine demand
planned_project_activity	Temporary or planned project activity
seasonal_usage	Expected seasonal pattern
incident_or_resilience_event	Cost increased due to incident, failover, DR, or recovery
deployment_or_architecture_change	Change in system design, deployment, or scaling behaviour
temporary_testing_or_migration	Short-term engineering activity
waste_or_inefficiency	Team accepts that optimisation is needed
incorrect_assignment	Wrong product, application, team, or cost centre mapping
unknown_needs_investigation	Team cannot currently explain the issue
data_quality_issue	Report data appears incorrect or incomplete

Example User Flow

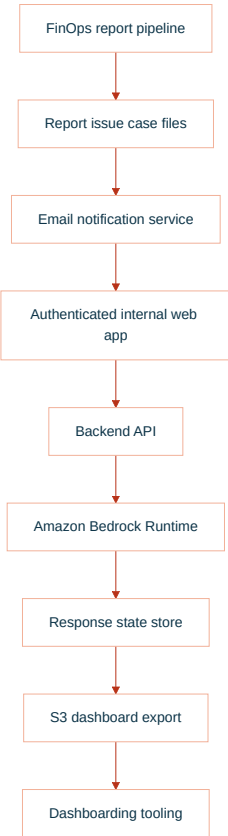


The final submit action is handled by deterministic backend logic, not by the model directly.

Example Reassignment Flow



Recommended Architecture



Component Responsibilities

Component	Responsibility
Report pipeline	Generates report issue case files
Email notification service	Sends team-specific deep links
Web frontend	Provides chat interface and issue view
Backend API	Performs authz, case retrieval, Bedrock invocation, validation, writes
Bedrock Runtime	Generates explanations, questions, summaries, and structured drafts
Response state store	Stores canonical response state
S3 export bucket/prefix	Provides dashboard-ready response data
Dashboard tooling	Displays response status, commentary, and reassignment state
Audit log	Records all views, submissions, model interactions, and writes

Recommended Storage Design

Preferred

Data	Recommended Store
Current case state	DynamoDB or RDS/Postgres
Conversation state	DynamoDB or RDS/Postgres
Structured justification	DynamoDB or RDS/Postgres
Dashboard export	S3
Analytics	Athena/Glue over S3
Long-term audit	CloudWatch Logs and/or S3

S3-Only Fallback

If the enterprise insists on S3 as the main integration point:

Data	Store
Current response	S3 object
Response history	Append-only S3 objects
Dashboard output	S3 partitioned data
Audit	CloudWatch and/or S3 logs

S3 should be treated as the dashboard/export layer where possible, not the primary transactional workflow store.

Example Data Model

Cost Issue Record

JSON

```
{
  "issue_id": "finops-2026-05-001234",
  "report_id": "monthly-cloud-report-2026-05",
  "report_period": "2026-05",
  "status": "awaiting_response",
  "assigned_team": "Payments Platform",
  "assigned_application": "Checkout API",
  "assigned_product": "Payments",
  "aws_account_id": "123456789012",
  "service": "AmazonEC2",
  "cost_delta": 18500.25,
  "cost_delta_percentage": 32.4,
  "currency": "GBP",
  "summary": "EC2 spend increased materially versus prior month.",
  "response_due_date": "2026-06-07"
}
```

Justification Response

JSON

```
{
  "issue_id": "finops-2026-05-001234",
  "response_id": "resp-001",
  "response_type": "justification",
  "submitted_by": "user@company.com",
  "submitted_at": "2026-05-21T14:30:00Z",
  "team": "Payments Platform",
  "justification_category": "expected_business_growth",
  "business_reason": "Traffic increased due to new merchant onboarding.",
  "technical_reason": "Additional ECS capacity was required for the checkout service.",
  "expected_to_continue": true,
  "expected_duration": "ongoing",
  "requires_finops_review": false,
  "confidence": "medium",
  "supporting_links": [
    "https://internal/change/CH612345"
  ],
  "ai_summary": "The team states the increase is expected and linked to merchant onboarding and increased checkout demand."
}
```

Reassignment Request

JSON

```
{
  "issue_id": "finops-2026-05-001234",
  "response_type": "reassignment_request",
  "submitted_by": "user@company.com",
  "submitted_at": "2026-05-21T14:35:00Z",
  "current_assignment": {
    "product": "Payments",
    "application": "Checkout API",
    "team": "Payments Platform"
  },
  "proposed_assignment": {
    "product": "Customer Analytics",
    "application": "Reporting Pipeline",
    "team": "Data Platform"
  },
  "reason": "The resources are tagged with the old Payments cost centre but are used by the reporting pipeline.",
  "requires_catalog_update": true,
  "status": "pending_review"
}
```

Dashboard Export Record

JSON

```
{
  "report_period": "2026-05",
  "issue_id": "finops-2026-05-001234",
  "product": "Payments",
  "application": "Checkout API",
  "team": "Payments Platform",
  "service": "AmazonEC2",
  "cost_delta": 18500.25,
  "status": "justified",
  "response_type": "expected_business_growth",
  "response_summary": "Increase accepted as expected business growth due to new merchant onboarding.",
  "submitted_by": "user@company.com",
  "submitted_at": "2026-05-21T14:30:00Z",
  "requires_finops_review": false,
  "requires_catalog_update": false
}
```

Recommended S3 Layout

TEXT

```
s3://<bucket>/report-issues/report_period=2026-05/issue_id=...json
s3://<bucket>/responses-current/report_period=2026-05/issue_id=...json
s3://<bucket>/responses-history/report_period=2026-05/issue_id=.../event_ts=...json
s3://<bucket>/dashboard-export/report_period=2026-05/team=.../part-....parquet
```

For dashboarding, Parquet is preferable to many individual JSON files if the data will be queried through Athena, QuickSight, Power BI, Tableau, or similar tooling.

AI Responsibilities

The AI assistant should help with:

Function	Description
Issue explanation	Explain the financial report item in plain English
Guided response	Ask the user for the minimum useful justification
Classification	Convert free text into controlled categories
Reassignment support	Suggest matching catalogue entries where appropriate
Dashboard summary	Produce concise finance-friendly commentary
FAQ support	Explain methodology, assignment, and terminology

The AI should not:

Prohibited Behaviour	Reason
Directly write to S3	Backend must validate writes
Directly update the technology catalogue	Reassignment requires review
Directly modify infrastructure	Out of scope and high risk
Invent causes without evidence	Reduces trust and increases risk
Access unrelated team data	Violates least privilege
Act without user confirmation	Workflow must be human-led

Prompt Context Pattern

The model should receive a compact case file, not raw CUR data.

Example:

JSON

```
{
  "task": "Assist the user in responding to a FinOps report issue.",
  "issue": {
    "issue_id": "finops-2026-05-001234",
    "summary": "EC2 spend increased by £18.5k month-on-month.",
    "current_assignment": {
      "product": "Payments",
      "application": "Checkout API",
      "team": "Payments Platform"
    },
  },
  "detected_drivers": [
    "c7g.4xlarge running hours increased 41%",
    "EBS gp3 storage increased 19%",
    "Autoscaling minimum capacity appears to have increased"
  ],
  "known_events": [
    "Change CH612345: raised production capacity",
    "Release 2026.05.14: merchant onboarding changes"
  ]
},
  "allowed_actions": [
    "submit_justification",
    "request_reassignment",
    "mark_needs_investigation",
    "attach_evidence"
  ],
  "policy": {
    "do_not_claim_certainty_without_evidence": true,
    "ask_for_confirmation_before_submission": true,
    "do_not_update_catalog_directly": true
  }
}
```

Guardrails and Validation

Controls should exist both at the application level and, where appropriate, through Bedrock Guardrails.

Control	Implementation
User must authenticate	Enterprise SSO
User must be authorised for issue	Application RBAC/ABAC
AI only receives case-specific context	Backend context builder
No unsupported certainty	Prompt rules and output validation
No direct catalogue mutation	Workflow design and IAM restrictions
Confirmation before submit	Application UX
Structured output schema	JSON schema validation
Sensitive data redaction	Application preprocessing and/or guardrails
Audit trail	Application audit events and logs
Prompt/output retention decision	Security-approved logging policy

AWS Enablement Checklist

The following sections list the AWS and enterprise enablement items that should be explicitly requested. In this organisation, do not assume implied dependencies will be granted automatically.

1. AWS Account and Region

Requirement	Request	Notes
AWS account	Dedicated workload account or approved shared account	Prefer separate non-prod and prod
Region	Approved Bedrock-supported region	Must satisfy data residency requirements
Environment separation	Dev/test/prod or equivalent	At minimum non-prod and prod
SCP exceptions	Allow approved services and actions	Many enterprises block AI and Marketplace services by default
Mandatory tags	Apply enterprise tagging standard	Application, owner, cost centre, environment, data classification

2. Bedrock Access

Requirement	Exact Ask
Enable Amazon Bedrock	In selected account and region
Model approval	Approve one text model for MVP
Model invocation	Allow runtime role to invoke approved model
Streaming	Decide whether streaming responses are allowed
Marketplace/legal review	Required if using third-party model provider
Guardrails	Create/approve guardrail if required
Invocation logging	Enable if required by security

Minimum runtime actions:

```
JSON

{
  "Effect": "Allow",
  "Action": [
    "bedrock:InvokeModel",
    "bedrock:InvokeModelWithResponseStream"
  ],
  "Resource": [
    "arn:aws:bedrock:<region>:foundation-model/<approved-model-id>"
  ]
}
```

If using the Bedrock Converse API, include equivalent model invocation permissions as required by the selected SDK/API pattern.

3. Bedrock Network Connectivity

For a no-internet environment, request Bedrock PrivateLink access.

Endpoint	Required?	Purpose
<code>com.amazonaws.<region>.bedrock-runtime</code>	Yes	Model invocation
<code>com.amazonaws.<region>.bedrock</code>	Maybe	Control-plane/list/configuration operations
<code>com.amazonaws.<region>.bedrock-agent-runtime</code>	Only if using Agents	Invoke Bedrock Agent
<code>com.amazonaws.<region>.bedrock-agent</code>	Only if managing Agents	Agent build/configuration

For MVP, request only what is needed:

TEXT
<code>com.amazonaws.<region>.bedrock-runtime</code>

Network details:

Item	Required
Interface VPC endpoint	Bedrock Runtime
Private DNS	Enabled
Endpoint security group	Allow TCP 443 from app runtime security group
App security group egress	Allow TCP 443 to endpoint security group
Endpoint policy	Restrict to approved principals/actions/models where possible
DNS resolution	Must work from app runtime, bastion, and CI runner if applicable
Route dependency	No NAT or public internet dependency

4. Runtime Hosting

Choose one runtime pattern.

Recommended Options

Runtime	Good For	Notes
ECS Fargate	Internal web app/API	Good balance of control and simplicity
Lambda + API Gateway	Lightweight MVP	May be sufficient for simple workflows
Existing internal app platform	Enterprise alignment	Often easiest to approve
EKS	Existing Kubernetes platform	More dependencies and operational burden

ECS Fargate Requirements

Resource	Required
ECS cluster	Existing or new
ECS service	Runs the backend
Task execution role	Pull image and write logs
Task runtime role	Bedrock/data/log permissions
ECR repository	Store container image
Internal ALB or approved ingress	Internal HTTPS access
Security groups	App ingress and endpoint egress
Private subnets	Runtime placement
CloudWatch Logs	Container logs
KMS	Secrets/log/data encryption
Secrets Manager	App secrets
CI/CD deploy role	Push image and update service

Lambda Requirements

Resource	Required
Lambda function	Backend/API handler
Lambda execution role	Bedrock/data/log permissions
API Gateway or internal ALB	Web/API entry point
VPC configuration	Required if accessing private endpoints/data stores
Lambda security group	Egress to VPC endpoints
CloudWatch Logs	Function logs
KMS	Environment variables/secrets
Deployment role	CI/CD updates

5. User Access and Authentication

The email link is not the security boundary.

Requirement	Ask
Enterprise SSO	OIDC/SAML integration with Entra ID, Okta, Ping, or equivalent
User identity claims	Email, name, user ID
Group/team claims	Required for RBAC/ABAC
Application roles	Responder, FinOps reviewer, catalogue reviewer, admin
Session management	Approved session lifetime and logout behaviour
Internal URL allowlist	App accessible from corporate network, VPN, or ZTNA
Deep link format	Link includes report ID and issue ID only
Authorisation	App checks user can view/respond to issue

Example link:

TEXT

https://finops-response.internal.company/report/monthly-2026-05/issue/finops-001234

Application authorisation rule:

TEXT

User is authenticated
AND user belongs to assigned team OR FinOps admin group
AND issue is open
AND requested action is allowed for user role

6. Application Data Store

Preferred Design

Data	Store
Case state	DynamoDB or RDS/Postgres
Conversation state	DynamoDB or RDS/Postgres
Audit events	DynamoDB/RDS plus log export
Dashboard output	S3
Analytics	Athena over S3

If S3-Based

Data	Store
Current response	S3 object
Response history	Append-only S3 objects
Dashboard output	S3 partitioned data
Audit	CloudWatch and/or S3 logs

7. S3 Buckets and Prefixes

Bucket / Prefix	Access Pattern
report-issues/*	App reads issue context
responses-current/*	App writes/reads current response state
responses-history/*	App writes append-only history
dashboard-export/*	App writes dashboard-ready data
ai-audit-logs/*	App/security audit logging if S3-backed
bedrock-invocation-logs/*	Bedrock invocation logs if enabled

Example app runtime permissions:

JSON

```
{
  "Effect": "Allow",
  "Action": [
    "s3:GetObject"
  ],
  "Resource": [
    "arn:aws:s3:::<bucket>/report-issues/*"
  ]
}
```

JSON

```
{
  "Effect": "Allow",
  "Action": [
    "s3:PutObject",
    "s3:GetObject"
  ],
  "Resource": [
    "arn:aws:s3:::<bucket>/dashboard-export/*",
    "arn:aws:s3:::<bucket>/responses-current/*",
    "arn:aws:s3:::<bucket>/responses-history/*"
  ]
}
```

Do not forget bucket-level listing if the application needs it:

JSON

```
{
  "Effect": "Allow",
  "Action": [
    "s3:ListBucket"
  ],
  "Resource": [
    "arn:aws:s3:::<bucket>"
  ],
  "Condition": {
    "StringLike": {
      "s3:prefix": [
        "report-issues/*",
        "responses-current/*",
        "responses-history/*",
        "dashboard-export/*"
      ]
    }
  }
}
```

8. S3 Event Notifications

Only required if downstream processing should trigger when response/export objects are written.

Item	Required If Event-Driven
S3 notification configuration	ObjectCreated events on response/export prefix
Destination	SQS, SNS, Lambda, or EventBridge
Destination resource policy	Allow S3 bucket to publish/invoke
Lambda/SQS permissions	If consuming events
Dead-letter queue	Recommended
Event filtering	Prefix/suffix filters

Important implied dependency:

TEXT

Creating the S3 notification is not enough. The destination must also allow the bucket/service to publish, invoke, or send messages.

9. KMS

Assume customer-managed KMS keys are required.

Principal	Required KMS Permissions
App runtime role	<code>kms:Decrypt</code> , <code>kms:Encrypt</code> , <code>kms:GenerateDataKey</code> for app data/export/log keys
Data pipeline role	Same for dashboard export and issue data
CloudWatch Logs service	Key policy permission if log groups are encrypted
Bedrock logging service	Key/bucket policy support if using encrypted S3 destination
Security audit role	<code>kms:Decrypt</code> where audit review is permitted
Dashboard reader role	<code>kms:Decrypt</code> for dashboard export data

Typical runtime KMS permissions:

```
JSON

{
  "Effect": "Allow",
  "Action": [
    "kms:Decrypt",
    "kms:Encrypt",
    "kms:GenerateDataKey"
  ],
  "Resource": [
    "arn:aws:kms:<region>:<account-id>:key/<key-id>"
  ]
}
```

For read-only issue data, only `kms:Decrypt` may be required.

10. Logging and Audit

Log / Audit Item	Required
Application access log	Who viewed which issue
Business action audit	Who submitted what, when, against which issue
Model interaction metadata	Model ID, prompt template version, response ID, timestamp
Prompt/output logging	Depends on security policy
CloudTrail	AWS API activity
CloudWatch Logs	App/runtime logs
Bedrock invocation logging	If required
S3 data events	If required by security

Decisions required from security:

Decision	Options
Log prompts?	Yes / no / redacted only
Log model outputs?	Yes / no / redacted only
Retention	30 / 90 / 365 days or enterprise standard
Access	Security only / app team / FinOps
Data classification	Internal / confidential
Redaction	Required before logging or not

11. CloudWatch Logs

If using ECS, Lambda, API Gateway, or similar runtime components, ask for:

Permission / Resource	Purpose
Pre-created log groups or permission to create	App logs
<code>logs:CreateLogStream</code>	Runtime writes log stream
<code>logs:PutLogEvents</code>	Runtime writes events
KMS key for log group	If logs are encrypted
Retention policy	Avoid indefinite retention
Metric filters/alarms	Optional operational monitoring

Example:

```
JSON

{
  "Effect": "Allow",
  "Action": [
    "logs:CreateLogStream",
    "logs:PutLogEvents"
  ],
  "Resource": [
    "arn:aws:logs:<region>:<account-id>:log-group:/aws/finops-response/*"
  ]
}
```

12. Secrets Manager

Needed if the app integrates with SSO, Jira, ServiceNow, catalogue APIs, dashboard APIs, or other internal services.

Item	Required
Secret ARNs	Specific secrets only
Runtime role permission	<code>secretsmanager:GetSecretValue</code>
KMS decrypt	For secrets key
Rotation owner	Defined owner and process
No plaintext environment variables	Security requirement

Example:

```
JSON

{
  "Effect": "Allow",
  "Action": [
    "secretsmanager:GetSecretValue"
  ],
  "Resource": [
    "arn:aws:secretsmanager:<region>:<account-id>:secret:finops-response/*"
  ]
}
```

13. Report Issue Context Access

The app should read a pre-built case file for each report issue.

Case file should contain only what is needed:

Field Category	Example
Issue ID	<code>finops-2026-05-001234</code>
Report period	<code>2026-05</code>
Current owner	Product/application/team
Cost movement	Amount, percentage, service
Drivers	Service/resource deltas
Evidence links	Query IDs, dashboard links, change records
Allowed responders	Team/group IDs
Current status	Awaiting response, justified, disputed
Response due date	Date

Ask for read access to the case file source, not raw CUR, unless absolutely necessary.

14. CUR, Athena, and Glue Access

Avoid direct CUR querying in the MVP if possible.

If the application must query FinOps data directly, it will need more than Athena access.

Service	Required Permissions
Athena	<code>athena:StartQueryExecution</code> , <code>athena:GetQueryExecution</code> , <code>athena:GetQueryResults</code> , <code>athena:StopQueryExecution</code>
Glue	<code>glue:GetDatabase</code> , <code>glue:GetDatabases</code> , <code>glue:GetTable</code> , <code>glue:GetTables</code> , <code>glue:GetPartition</code> , <code>glue:GetPartitions</code>
S3 source	<code>s3:GetObject</code> , <code>s3:ListBucket</code>
S3 output	<code>s3:PutObject</code> , <code>s3:GetObject</code> , <code>s3:ListBucket</code>
KMS	<code>kms:Decrypt</code> , <code>kms:GenerateDataKey</code>
Network	Athena, Glue, S3, STS, KMS endpoints if private
Workgroup	Access to approved Athena workgroup

Recommended MVP position:

TEXT

The report pipeline creates narrow case files. The response portal does not query raw CUR directly.

15. Technology Catalogue Access

For reassignment workflows, the app needs read access to the technology catalogue.

Need	Access
Lookup current assignment	Read product/application/team mapping
Search proposed assignment	Read catalogue entries
Validate product/application/team exists	Read catalogue
Submit reassignment request	Write request record, not direct catalogue mutation
Catalogue update	Separate approval workflow

If the catalogue is not AWS-hosted, explicitly request:

Item	Ask
Internal DNS name	e.g. <code>catalogue.internal.company</code>
Network route	App subnet to catalogue API
Port	TCP 443
Auth method	OAuth2 client credentials, mTLS, service account, or equivalent
Secret storage	Secrets Manager
Firewall rule	Explicit allow from app runtime
Certificate trust	Internal CA bundle if needed

16. Email Notification

Preferred Option: Existing Enterprise Email System

Requirement	Need
Secure deep link support	Link to issue page
Approved sender identity	FinOps mailbox or service account
Template approval	If required
Link routing	Internal URL accessible to intended users

Alternative Option: AWS SES

Requirement	Need
SES enabled in region	Send emails
Verified domain/sender	Required
IAM permission	<code>ses:SendEmail</code> , <code>ses:SendRawEmail</code>
Bounce/complaint handling	SNS/SQS if required
Email security approval	SPF, DKIM, DMARC alignment

In a locked-down enterprise, using the existing notification platform is usually easier than introducing SES.

17. Dashboard Integration

Define exactly how dashboard tooling reads the response output.

Pattern	Access Needed
QuickSight reads S3/Athena	QuickSight role needs S3, Athena, Glue, KMS
Power BI reads via gateway	Gateway/service account needs read path
Tableau reads Athena	Athena/ODBC/JDBC/network/KMS access
Internal dashboard reads API	API auth and network route
Existing FinOps dashboard consumes S3	Bucket/prefix read and KMS decrypt

For S3/Athena dashboarding:

Permission	Principal
<code>s3:GetObject</code> on dashboard export prefix	Dashboard role
<code>s3:ListBucket</code> with prefix condition	Dashboard role
<code>kms:Decrypt</code>	Dashboard role
Athena query permissions	Dashboard role, if querying
Glue catalogue read permissions	Dashboard role, if querying

18. Private Connectivity Matrix

From	To	Port	Protocol	Purpose
User browser	Internal app URL / ALB	443	HTTPS	Use chat portal
App runtime	Bedrock Runtime VPC endpoint	443	HTTPS	Invoke approved model
App runtime	S3 endpoint	443	HTTPS	Read/write report, response, export objects
App runtime	KMS endpoint	443	HTTPS	Encrypt/decrypt data
App runtime	CloudWatch Logs endpoint	443	HTTPS	Write logs
App runtime	Secrets Manager endpoint	443	HTTPS	Read app secrets
App runtime	DynamoDB endpoint	443	HTTPS	Read/write state, if used
App runtime	STS endpoint	443	HTTPS	Assume role, if cross-account
App runtime	Technology catalogue API	443	HTTPS	Read/validate assignment
App runtime	SSO/OIDC provider	443	HTTPS	Auth/token validation, if needed
CI runner	ECR API and Docker endpoints	443	HTTPS	Push/pull container
CI runner	ECS/Lambda/API deployment endpoints	443	HTTPS	Deploy app
Dashboard tool	S3/Athena/Glue/KMS	443	HTTPS/JDBC	Read exports

19. VPC Endpoints Beyond Bedrock

In a no-internet AWS environment, the workload may need the following endpoints depending on selected architecture:

Endpoint	Required If
S3 gateway/interface endpoint	Reading/writing S3
KMS interface endpoint	Using customer-managed keys
CloudWatch Logs endpoint	Writing runtime logs
Secrets Manager endpoint	Reading secrets
STS endpoint	Assuming roles
ECR API endpoint	ECS image pulls or CI/CD image pushes
ECR Docker endpoint	ECS image pulls
CloudWatch endpoint	Metrics and alarms
DynamoDB gateway endpoint	DynamoDB state store
Athena endpoint	Direct Athena queries
Glue endpoint	Data catalogue access
Lambda endpoint	Lambda deploy/invoke
ECS endpoint	ECS deploy/runtime operations
ELB endpoint	Deployment/ops if needed
X-Ray endpoint	Tracing if used

Important statement for security/platform teams:

TEXT

Requesting Bedrock PrivateLink alone is insufficient. The workload also needs private connectivity to every dependency it calls, including KMS, S3, CloudWatch Logs, Secrets Manager, STS, and the selected state store.

20. Cross-Account Access

If report data, Bedrock runtime, S3 export, dashboarding, and logs are in different AWS accounts, explicitly request cross-account access.

Cross-Account Path	Required
App account to FinOps data account	Assume role or bucket policy
App account to dashboard bucket account	PutObject permission
Dashboard account to export bucket	Read permission
Security account to logs	Read permission
CI/CD account to app account	Deploy role
Billing/management account to data export account	CUR/Data Exports access

Cross-account dependencies:

TEXT

sts:AssumeRole
Trust policy from target role to source role
External ID if required
SCP allowlist
S3 bucket policy allowing cross-account principal
KMS key policy allowing cross-account principal

Do not forget KMS. Cross-account S3 access often fails because the bucket policy is correct but the KMS key policy is not.

21. IAM Role Catalogue

Role	Used By	Needs
<code>FinOpsResponseAppRuntimeRole</code>	Backend app	Bedrock invoke, read issue data, write response/export, logs, KMS
<code>FinOpsResponseDeployRole</code>	CI/CD	Deploy app infra, update ECS/Lambda, read/write ECR
<code>FinOpsResponseTaskExecutionRole</code>	ECS only	Pull ECR image, write logs
<code>FinOpsResponseDataExportRole</code>	App or pipeline	Write dashboard S3 export
<code>FinOpsResponseDashboardReadRole</code>	Dashboard tooling	Read export data and decrypt
<code>FinOpsResponseAuditReadRole</code>	Security/FinOps audit	Read logs/audit records
<code>FinOpsResponseDeveloperRole</code>	Developers via bastion/SSO	Read non-prod logs/config, limited dev invoke
<code>FinOpsResponseCatalogueReadRole</code>	App, if AWS-hosted catalogue	Read catalogue data
<code>FinOpsResponseBedrockLoggingRole</code>	Bedrock/log delivery if required	Write invocation logs

Key design rule:

TEXT
No human user receives direct S3 write access to submit responses. All writes go through the application.

22. Runtime Role Permissions

For the MVP, the app runtime role may need:

Service	Actions
Bedrock	<code>bedrock:InvokeModel</code> , <code>bedrock:InvokeModelWithResponseStream</code>
S3 read issue context	<code>s3:GetObject</code> , limited <code>s3:ListBucket</code>
S3 write export	<code>s3:PutObject</code> , maybe <code>s3:GetObject</code>
DynamoDB, if used	<code>dynamodb:GetItem</code> , <code>dynamodb:PutItem</code> , <code>dynamodb:UpdateItem</code> , <code>dynamodb:Query</code>
KMS	<code>kms:Decrypt</code> , <code>kms:Encrypt</code> , <code>kms:GenerateDataKey</code>
CloudWatch Logs	<code>logs:CreateLogStream</code> , <code>logs:PutLogEvents</code>
Secrets Manager	<code>secretsmanager:GetSecretValue</code>
STS	<code>sts:AssumeRole</code> , only if cross-account
Technology catalogue	Usually non-AWS API auth/secret, or AWS permissions if hosted on AWS

Notably absent unless explicitly required:

TEXT
<code>ec2:*</code> <code>rds:*</code> <code>iam:*</code> <code>organizations:*</code> <code>ce:*</code> <code>cur:*</code> <code>pricing:*</code> <code>config:*</code> <code>cloudformation:*</code>

23. Bedrock Logging Requirements

If enabling Bedrock invocation logging, treat it as a separate security decision.

Item	Required
Bedrock invocation logging configuration	Account/region-level config
S3 log bucket	Same region/account constraints may apply depending on configuration
S3 bucket policy	Allow Bedrock log delivery
CloudWatch log group	If CloudWatch destination is used
IAM permissions for admin/config role	Configure logging
KMS permissions	If logs are encrypted
Retention	Defined
Access policy	Who can read prompt/output logs

Prompt and output logs may contain financial, organisational, or user-entered information. They should be treated as sensitive unless security states otherwise.

24. Data Classification Questions

Security and governance should answer these before production use.

Question	Required Decision
Can report issue data be sent to Bedrock?	Yes/no and classification
Can user justifications be sent to Bedrock?	Yes/no
Can prompts be logged?	Yes/no/redacted
Can outputs be logged?	Yes/no/redacted
Can model output be stored in dashboard data?	Yes/no
Are account IDs/resource IDs sensitive?	Required treatment
Are team names/owner names sensitive?	Required treatment
Can Bedrock use third-party model providers?	Legal/security decision
Which region is allowed?	Data residency decision
How long are transcripts retained?	Retention decision

25. Minimum MVP Access Request

A tight MVP access request:

Category	MVP Ask
AWS account	Approved non-prod workload account
Runtime	ECS Fargate or Lambda in private subnet
Bedrock	Invoke one approved model in one region
Network	PrivateLink to Bedrock Runtime, S3, KMS, CloudWatch Logs, Secrets Manager, STS, state store
Auth	Enterprise SSO integration
Data read	Read-only report issue case files
Data write	Write controlled response records
Dashboard	Write S3 dashboard export prefix
Logs	CloudWatch app logs and business audit events
KMS	Encrypt all S3, logs, state, and secrets
CI/CD	Approved deploy path from internal runner
Bastion	Developer CLI/debug access in non-prod only

26. Explicit Non-Requirements for MVP

Not Required	Reason
Raw CUR access from the app	Case files are generated upstream
Live AWS infrastructure read access	App does not inspect infra directly
AWS infrastructure write access	App does not remediate or mutate infra
Direct technology catalogue mutation	Reassignment requests require review
Broad Bedrock model access	One approved model only
Autonomous agent tools	Backend owns all actions
Public internet egress	Private access only
Static access keys	SSO/roles only
Cross-team visibility	Scoped authorisation only

27. Write Path Summary

The system should only support the following write actions in MVP:

Action	Writes To	Approval / Confirmation
Submit justification	Response state and dashboard export	User confirmation
Mark needs investigation	Response state and dashboard export	User confirmation
Dispute assignment	Response state and dashboard export	User confirmation
Request reassignment	Reassignment request record	FinOps/catalogue review
Attach evidence link	Response state	User confirmation
Close response	Response state	FinOps/admin or authorised team

The system does not write to:

TEXT
AWS infrastructure
Billing configuration
Cost allocation tags
IAM
Technology catalogue directly
Production services

28. Security Enablement Matrix Template

Use this matrix when raising requests with platform/security teams.

ID	Component	Needs Access To	Direction	Protocol / Port	Principal	IAM Action	Resource Scope	Network Dependency	KMS Dependency
NET-001	App runtime	Bedrock Runtime VPC Endpoint	Outbound	HTTPS 443	App runtime SG / role	<code>bedrock:InvokeModel</code>	Approved model ARN	PrivateLink endpoint, SG, private DNS	None
IAM-001	App runtime role	Bedrock model	N/A	N/A	<code>FinOpsResponseAppRuntimeRole</code>	<code>bedrock:InvokeModel</code>	Approved model ARN	Bedrock endpoint	None
S3-001	App runtime	Report issue case files	Read	HTTPS 443	<code>FinOpsResponseAppRuntimeRole</code>	<code>s3:GetObject</code> , limited <code>s3:ListBucket</code>	<code>report-issues/*</code>	S3 endpoint	<code>kms:Decrypt</code>
S3-002	App runtime	Dashboard export prefix	Write	HTTPS 443	<code>FinOpsResponseAppRuntimeRole</code>	<code>s3:PutObject</code>	<code>dashboard-export/*</code>	S3 endpoint	<code>kms:Encrypt</code> , <code>kms:GenerateDataKe</code>
KMS-001	App runtime	Data KMS key	N/A	N/A	<code>FinOpsResponseAppRuntimeRole</code>	<code>kms:Decrypt</code> , <code>kms:Encrypt</code> , <code>kms:GenerateDataKey</code>	Approved key ARN	KMS endpoint	N/A
LOG-001	App runtime	CloudWatch Logs	Write	HTTPS 443	Task/Lambda role	<code>logs:CreateLogStream</code> , <code>logs:PutLogEvents</code>	App log group	Logs endpoint	KMS if encrypted
AUTH-001	User	SSO provider	Inbound/auth	HTTPS 443	User/browser/app	N/A	OIDC/SAML client	Internal route	N/A
CAT-001	App runtime	Technology catalogue API	Outbound	HTTPS 443	App service identity	API-specific	Catalogue read/search	Firewall/DNS route	Secret KMS decrypt
DASH-001	Dashboard role	S3 dashboard export	Read	HTTPS 443	Dashboard service role	<code>s3:GetObject</code> , <code>s3:ListBucket</code>	<code>dashboard-export/*</code>	S3/Athena route	<code>kms:Decrypt</code>
CICD-001	CI/CD runner	ECR and runtime deploy APIs	Outbound	HTTPS 443	Deploy role	Service-specific	App resources only	ECR/ECS/Lambda endpoints	KMS as needed
AUD-001	Security audit	Logs and audit records	Read	HTTPS 443	Audit role	Read-only log/data actions	Audit resources	Logs/S3 endpoint	<code>kms:Decrypt</code>

29. Key Security Statements

Use these statements in the design review:

TEXT
Bedrock does not receive broad AWS permissions.
TEXT
The model does not access AWS services directly.
TEXT
The backend retrieves a narrow case file for a specific report issue and sends only that context to Bedrock.
TEXT
All writes are performed by the backend after user confirmation, schema validation, and authorisation checks.
TEXT
The MVP does not change infrastructure, cost allocation tags, IAM, billing configuration, or the technology catalogue.
TEXT
The system writes only structured response records and dashboard exports.

TEXT
No human user receives direct S3 write access for submitting responses.

30. Suggested Delivery Plan

Phase	Scope	Outcome
Phase 1: Paper design	Architecture, access matrix, security review	Approval to build non-prod MVP
Phase 2: Non-prod MVP	One report type, one model, one team, synthetic or limited data	Validate workflow and controls
Phase 3: Pilot	Real report items for selected teams	Validate usefulness, audit, dashboard integration
Phase 4: Production hardening	Monitoring, logging, DR, retention, guardrails, review process	Production readiness
Phase 5: Expansion	More report types, catalogue workflow, richer insight	Broader FinOps adoption

31. Risks and Mitigations

Risk	Mitigation
Security rejects broad AI access	Narrow case-file design; Bedrock has no broad AWS access
Prompt/output logs contain sensitive data	Explicit logging policy, redaction, restricted access
AI gives unsupported explanation	Evidence-first prompting and confirmation workflow
Users submit poor justifications	Guided fields and controlled categories
Reassignment causes ownership disputes	Reassignment request workflow with review
Dashboard data becomes inconsistent	Structured schema and canonical state store
Cross-account KMS blocks access	Include KMS key policy requirements upfront
Private networking misses dependencies	Use full connectivity matrix
Model access blocked by SCP/legal	Include model approval and SCP exceptions in initial request

32. Final Recommendation

For the first version, keep the AWS and security ask intentionally narrow:

TEXT
One approved model. One private runtime. One report issue data source. One response data store. One S3 dashboard export prefix. One SSO integration. No raw CUR access. No live infrastructure API access. No catalogue mutation. No autonomous remediation.

This gives the organisation a controlled, auditable, low-blast-radius use case that demonstrates practical AI-native FinOps value without creating an unbounded AI security problem.